

10. The tables show the molecular formulae of some alkanes and alkenes.

Alkanes	Alkenes
CH_4	C_2H_4
C_2H_6	C_3H_6
C_3H_8	
C_4H_{10}	

- (a) The general formula for the alkene family is C_nH_{2n} . Give the general formula for the alkane family. [1]

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- (b) When alkanes and alkenes completely burn in air they form the same two products.
Give the chemical formulae for both products. [1]

..... and

- (c) Draw the structural formula for propene. [1]

- (d) Bromine water is used to distinguish alkenes from alkanes. Describe the colour **change** seen when bromine water is added to an alkene. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



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Key

Ar	Symbol	Name	Z
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